MAT110

Differentiation

Tangent Line:

2.1.1 DEFINITION Suppose that x_0 is in the domain of the function f. The ***tangent line*** to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is the line with equation

$$y - f(x_0) = m_{\tan}(x - x_0)$$

where

$$m_{\tan} = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \tag{1}$$

provided the limit exists. For simplicity, we will also call this the tangent line to $y = f(x)$ at x_0.

▶ Example 1 Use Definition 2.1.1 to find an equation for the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$, and confirm the result agrees with that obtained in Example 1 of Section 1.1.

Solution. Applying Formula (1) with $f(x) = x^2$ and $x_0 = 1$, we have

$$\begin{aligned} m_{\tan} &= \lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} \\ &= \lim_{x \to 1} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x \to 1} \frac{(x-1)(x+1)}{x - 1} = \lim_{x \to 1}(x + 1) = 2 \end{aligned}$$

Thus, the tangent line to $y = x^2$ at $(1, 1)$ has equation

$$y - 1 = 2(x - 1) \quad \text{or equivalently} \quad y = 2x - 1$$

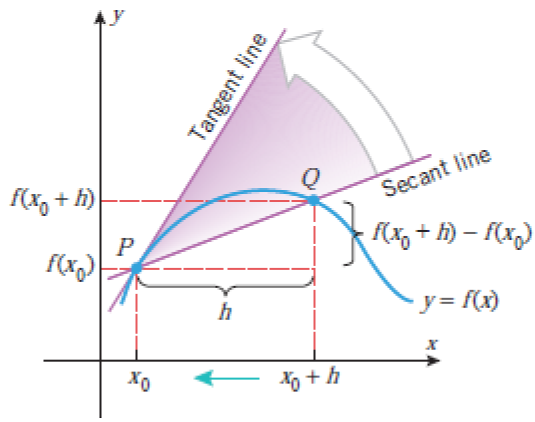
There is an alternative way of expressing Formula (1) that is commonly used. If we let h denote the difference

$$h = x - x_0$$

then the statement that $x \to x_0$ is equivalent to the statement $h \to 0$, so we can rewrite (1) in terms of x_0 and h as

$$m_{\tan} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \tag{2}$$

Figure 2.1.2 shows how Formula (2) expresses the slope of the tangent line as a limit of slopes of secant lines.



▶ **Figure 2.1.2**

▶ **Example 2** Compute the slope in Example 1 using Formula (2).

Solution. Applying Formula (2) with $f(x) = x^2$ and $x_0 = 1$, we obtain

$$
\begin{aligned}
m_{\tan} &= \lim_{h \to 0} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0} \frac{(1+h)^2 - 1^2}{h} \\
&= \lim_{h \to 0} \frac{1 + 2h + h^2 - 1}{h} = \lim_{h \to 0} (2 + h) = 2
\end{aligned}
$$

which agrees with the slope found in Example 1. ◀

DEFINITION OF THE DERIVATIVE FUNCTION

In the last section we showed that if the limit

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists, then it can be interpreted either as the slope of the tangent line to the curve $y = f(x)$ at $x = x_0$ or as the instantaneous rate of change of y with respect to x at $x = x_0$ [see Formulas (2) and (11) of that section]. This limit is so important that it has a special notation:

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \tag{1}$$

You can think of f' (read "f prime") as a function whose input is x_0 and whose output is the number $f'(x_0)$ that represents either the slope of the tangent line to $y = f(x)$ at $x = x_0$ or the instantaneous rate of change of y with respect to x at $x = x_0$. To emphasize this function point of view, we will replace x_0 by x in (1) and make the following definition.

2.2.1 DEFINITION The function f' defined by the formula

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} \tag{2}$$

is called the ***derivative of f with respect to x***. The domain of f' consists of all x in the domain of f for which the limit exists.

▶ Example 1 Find the derivative with respect to x of $f(x) = x^2$, and use it to find the equation of the tangent line to $y = x^2$ at $x = 2$.

Solution. It follows from (2) that

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \to 0} \frac{(x + h)^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \to 0} (2x + h) = 2x \end{aligned}$$

Thus, the slope of the tangent line to $y = x^2$ at $x = 2$ is $f'(2) = 4$. Since $y = 4$ if $x = 2$, the point-slope form of the tangent line is

$$y - 4 = 4(x - 2)$$

which we can rewrite in slope-intercept form as $y = 4x - 4$ (Figure 2.2.1). ◀

▶ **Example 6** The graph of $y = |x|$ in Figure 2.2.10 has a corner at $x = 0$, which implies that $f(x) = |x|$ is not differentiable at $x = 0$.

(a) Prove that $f(x) = |x|$ is not differentiable at $x = 0$ by showing that the limit in Definition 2.2.2 does not exist at $x = 0$.

Solution (a). From Formula (5) with $x_0 = 0$, the value of $f'(0)$, if it were to exist, would be given by

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{|h| - |0|}{h} = \lim_{h\to 0} \frac{|h|}{h} \qquad (6)$$

But

$$\frac{|h|}{h} = \begin{cases} 1, & h > 0 \\ -1, & h < 0 \end{cases}$$

so that

$$\lim_{h\to 0^-} \frac{|h|}{h} = -1 \quad \text{and} \quad \lim_{h\to 0^+} \frac{|h|}{h} = 1$$

Since these one-sided limits are not equal, the two-sided limit in (5) does not exist, and hence f is not differentiable at $x = 0$.

THE RELATIONSHIP BETWEEN DIFFERENTIABILITY AND CONTINUITY

We already know that functions are not differentiable at corner points and points of vertical tangency. The next theorem shows that functions are not differentiable at points of discontinuity. We will do this by proving that if f is differentiable at a point, then it must be continuous at that point.

2.2.3 THEOREM *If a function f is differentiable at x_0, then f is continuous at x_0.*

Techniques of Differentiation

Chain Rule

2.6.1 THEOREM (*The Chain Rule*) *If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x. Moreover, if*

$$y = f(g(x)) \quad \text{and} \quad u = g(x)$$

then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \qquad (1)$$

▶ Example 1 Find dy/dx if $y = \cos(x^3)$.

Solution. Let $u = x^3$ and express y as $y = \cos u$. Applying Formula (1) yields

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\
&= \frac{d}{du}[\cos u] \cdot \frac{d}{dx}[x^3] \\
&= (-\sin u) \cdot (3x^2) \\
&= (-\sin(x^3)) \cdot (3x^2) = -3x^2 \sin(x^3) \quad \blacktriangleleft
\end{aligned}$$

▶ Example 2 Find dw/dt if $w = \tan x$ and $x = 4t^3 + t$.

Solution. In this case the chain rule computations take the form

$$\begin{aligned}
\frac{dw}{dt} &= \frac{dw}{dx} \cdot \frac{dx}{dt} \\
&= \frac{d}{dx}[\tan x] \cdot \frac{d}{dt}[4t^3 + t] \\
&= (\sec^2 x) \cdot (12t^2 + 1) \\
&= [\sec^2(4t^3 + t)] \cdot (12t^2 + 1) = (12t^2 + 1)\sec^2(4t^3 + t) \quad \blacktriangleleft
\end{aligned}$$

AN ALTERNATIVE VERSION OF THE CHAIN RULE

Formula (1) for the chain rule can be unwieldy in some problems because it involves so many variables. As you become more comfortable with the chain rule, you may want to dispense with writing out the dependent variables by expressing (1) in the form

$$\frac{d}{dx}[f(g(x))] = (f \circ g)'(x) = f'(g(x))g'(x) \tag{2}$$

A convenient way to remember this formula is to call f the "outside function" and g the "inside function" in the composition $f(g(x))$ and then express (2) in words as:

> *The derivative of $f(g(x))$ is the derivative of the outside function evaluated at the inside function times the derivative of the inside function.*

$$\frac{d}{dx}[f(g(x))] = \underbrace{f'(g(x))}_{\text{Derivative of the outside function evaluated at the inside function}} \cdot \underbrace{g'(x)}_{\text{Derivative of the inside function}}$$

▸ **Example 5** Find

(a) $\dfrac{d}{dx}[\sin(2x)]$ (b) $\dfrac{d}{dx}[\tan(x^2+1)]$ (c) $\dfrac{d}{dx}\left[\sqrt{x^3+\csc x}\right]$

(d) $\dfrac{d}{dx}[x^2-x+2]^{3/4}$ (e) $\dfrac{d}{dx}\left[(1+x^5\cot x)^{-8}\right]$

Solution (a). Taking $u = 2x$ in the generalized derivative formula for $\sin u$ yields

$$\frac{d}{dx}[\sin(2x)] = \frac{d}{dx}[\sin u] = \cos u\frac{du}{dx} = \cos 2x \cdot \frac{d}{dx}[2x] = \cos 2x \cdot 2 = 2\cos 2x$$

Solution (b). Taking $u = x^2+1$ in the generalized derivative formula for $\tan u$ yields

$$\begin{aligned}\frac{d}{dx}[\tan(x^2+1)] &= \frac{d}{dx}[\tan u] = \sec^2 u\frac{du}{dx}\\ &= \sec^2(x^2+1)\cdot\frac{d}{dx}[x^2+1] = \sec^2(x^2+1)\cdot 2x\\ &= 2x\sec^2(x^2+1)\end{aligned}$$

Solution (c). Taking $u = x^3+\csc x$ in the generalized derivative formula for $\sqrt{u}$ yields

$$\begin{aligned}\frac{d}{dx}\left[\sqrt{x^3+\csc x}\right] &= \frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}}\frac{du}{dx} = \frac{1}{2\sqrt{x^3+\csc x}}\cdot\frac{d}{dx}[x^3+\csc x]\\ &= \frac{1}{2\sqrt{x^3+\csc x}}\cdot(3x^2-\csc x\cot x) = \frac{3x^2-\csc x\cot x}{2\sqrt{x^3+\csc x}}\end{aligned}$$

Solution (d). Taking $u = x^2-x+2$ in the generalized derivative formula for $u^{3/4}$ yields

$$\begin{aligned}\frac{d}{dx}[x^2-x+2]^{3/4} &= \frac{d}{dx}[u^{3/4}] = \frac{3}{4}u^{-1/4}\frac{du}{dx}\\ &= \frac{3}{4}(x^2-x+2)^{-1/4}\cdot\frac{d}{dx}[x^2-x+2]\\ &= \frac{3}{4}(x^2-x+2)^{-1/4}(2x-1)\end{aligned}$$

Solution (e). Taking $u = 1+x^5\cot x$ in the generalized derivative formula for u^{-8} yields

$$\begin{aligned}\frac{d}{dx}\left[(1+x^5\cot x)^{-8}\right] &= \frac{d}{dx}[u^{-8}] = -8u^{-9}\frac{du}{dx}\\ &= -8(1+x^5\cot x)^{-9}\cdot\frac{d}{dx}[1+x^5\cot x]\\ &= -8(1+x^5\cot x)^{-9}\cdot\left[x^5(-\csc^2 x)+5x^4\cot x\right]\\ &= (8x^5\csc^2 x-40x^4\cot x)(1+x^5\cot x)^{-9}\end{aligned}$$ ◂

Implicit Differentiation

FUNCTIONS DEFINED EXPLICITLY AND IMPLICITLY

An equation of the form $y = f(x)$ is said to define y ***explicitly*** as a function of x because the variable y appears alone on one side of the equation and does not appear at all on the other side. However, sometimes functions are defined by equations in which y is not alone on one side; for example, the equation

$$yx + y + 1 = x \tag{1}$$

is not of the form $y = f(x)$, but it still defines y as a function of x since it can be rewritten as

$$y = \frac{x-1}{x+1}$$

Thus, we say that (1) defines y ***implicitly*** as a function of x, the function being

$$f(x) = \frac{x-1}{x+1}$$

IMPLICIT DIFFERENTIATION

In general, it is not necessary to solve an equation for y in terms of x in order to differentiate the functions defined implicitly by the equation. To illustrate this, let us consider the simple equation

$$xy = 1 \tag{5}$$

One way to find dy/dx is to rewrite this equation as

$$y = \frac{1}{x} \tag{6}$$

from which it follows that

$$\frac{dy}{dx} = -\frac{1}{x^2} \tag{7}$$

Another way to obtain this derivative is to differentiate both sides of (5) *before* solving for y in terms of x, treating y as a (temporarily unspecified) differentiable function of x. With this approach we obtain

$$\frac{d}{dx}[xy] = \frac{d}{dx}[1]$$

$$x\frac{d}{dx}[y] + y\frac{d}{dx}[x] = 0$$

$$x\frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

If we now substitute (6) into the last expression, we obtain

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

which agrees with Equation (7). This method of obtaining derivatives is called ***implicit differentiation***.

▶ **Example 2** Use implicit differentiation to find dy/dx if $5y^2 + \sin y = x^2$.

$$\frac{d}{dx}[5y^2 + \sin y] = \frac{d}{dx}[x^2]$$

$$5\frac{d}{dx}[y^2] + \frac{d}{dx}[\sin y] = 2x$$

$$5\left(2y\frac{dy}{dx}\right) + (\cos y)\frac{dy}{dx} = 2x$$

The chain rule was used here because y is a function of x.

$$10y\frac{dy}{dx} + (\cos y)\frac{dy}{dx} = 2x$$

▶ **Example 3** Use implicit differentiation to find d^2y/dx^2 if $4x^2 - 2y^2 = 9$.

Solution. Differentiating both sides of $4x^2 - 2y^2 = 9$ with respect to x yields

$$8x - 4y\frac{dy}{dx} = 0$$

from which we obtain

$$\frac{dy}{dx} = \frac{2x}{y} \tag{9}$$

Differentiating both sides of (9) yields

$$\frac{d^2y}{dx^2} = \frac{(y)(2) - (2x)(dy/dx)}{y^2} \tag{10}$$

Substituting (9) into (10) and simplifying using the original equation, we obtain

$$\frac{d^2y}{dx^2} = \frac{2y - 2x(2x/y)}{y^2} = \frac{2y^2 - 4x^2}{y^3} = -\frac{9}{y^3}$$ ◀

LOGARITHMIC DIFFERENTIATION

We now consider a technique called ***logarithmic differentiation*** that is useful for differentiating functions that are composed of products, quotients, and powers.

▶ **Example 5** The derivative of

$$y = \frac{x^2\sqrt[3]{7x - 14}}{(1 + x^2)^4} \tag{7}$$

is messy to calculate directly. However, if we first take the natural logarithm of both sides and then use its properties, we can write

$$\ln y = 2\ln x + \tfrac{1}{3}\ln(7x - 14) - 4\ln(1 + x^2)$$

Differentiating both sides with respect to x yields

$$\frac{1}{y}\frac{dy}{dx} = \frac{2}{x} + \frac{7/3}{7x - 14} - \frac{8x}{1 + x^2}$$

Thus, on solving for dy/dx and using (7) we obtain

$$\frac{dy}{dx} = \frac{x^2\sqrt[3]{7x - 14}}{(1 + x^2)^4}\left[\frac{2}{x} + \frac{1}{3x - 6} - \frac{8x}{1 + x^2}\right]$$ ◀

► **Example 4** Use logarithmic differentiation to find $\frac{d}{dx}[(x^2+1)^{\sin x}]$.

Solution. Setting $y = (x^2+1)^{\sin x}$ we have

$$\ln y = \ln[(x^2+1)^{\sin x}] = (\sin x)\ln(x^2+1)$$

Differentiating both sides with respect to x yields

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= \frac{d}{dx}[(\sin x)\ln(x^2+1)] \\ &= (\sin x)\frac{1}{x^2+1}(2x) + (\cos x)\ln(x^2+1)\end{aligned}$$

Thus,

$$\begin{aligned}\frac{dy}{dx} &= y\left[\frac{2x\sin x}{x^2+1} + (\cos x)\ln(x^2+1)\right] \\ &= (x^2+1)^{\sin x}\left[\frac{2x\sin x}{x^2+1} + (\cos x)\ln(x^2+1)\right]\end{aligned}$$ ◄

Derivatives of Inverse Trigonometric Functions

formulas, each of which is valid on the natural domain of the function that multiplies du/dx.

$$\frac{d}{dx}[\sin^{-1} u] = \frac{1}{\sqrt{1-u^2}}\frac{du}{dx} \qquad \frac{d}{dx}[\cos^{-1} u] = -\frac{1}{\sqrt{1-u^2}}\frac{du}{dx} \qquad (9\text{–}10)$$

$$\frac{d}{dx}[\tan^{-1} u] = \frac{1}{1+u^2}\frac{du}{dx} \qquad \frac{d}{dx}[\cot^{-1} u] = -\frac{1}{1+u^2}\frac{du}{dx} \qquad (11\text{–}12)$$

$$\frac{d}{dx}[\sec^{-1} u] = \frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx} \qquad \frac{d}{dx}[\csc^{-1} u] = -\frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx} \qquad (13\text{–}14)$$

Example 5 Find dy/dx if

(a) $y = \sin^{-1}(x^3)$ (b) $y = \sec^{-1}(e^x)$

Solution (a). From (9)

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x^3)^2}}(3x^2) = \frac{3x^2}{\sqrt{1-x^6}}$$

Solution (b). From (13)

$$\frac{dy}{dx} = \frac{1}{e^x\sqrt{(e^x)^2-1}}(e^x) = \frac{1}{\sqrt{e^{2x}-1}}$$

Parametric Differentiation

The parametric Definition of Curve

In the first example below we shall show how the x and y coordinates of points on a curve can be defined in terms of a third variable, t, the **parameter**.

Example

Consider the parametric equations

$$x = \cos t \qquad y = \sin t \qquad \text{for } 0 \le t \le 2\pi \qquad (1)$$

Note how both x and y are given in terms of the third variable t.

3. Differentiation of a function defined parametrically

It is often necessary to find the rate of change of a function defined parametrically; that is, we want to calculate $\frac{dy}{dx}$. The following example will show how this is achieved.

Example

Suppose we wish to find $\frac{dy}{dx}$ when $x = \cos t$ and $y = \sin t$.

We differentiate both x and y with respect to the parameter, t:

$$\frac{dx}{dt} = -\sin t \qquad\qquad \frac{dy}{dt} = \cos t$$

From the chain rule we know that

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

so that, by rearrangement

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \qquad \text{provided } \frac{dx}{dt} \text{ is not equal to } 0$$

So, in this case

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\cos t}{-\sin t} = -\cot t$$

Example

Suppose we wish to find $\frac{dy}{dx}$ when $x = t^3 - t$ and $y = 4 - t^2$.

$$x = t^3 - t \qquad\qquad y = 4 - t^2$$

$$\frac{dx}{dt} = 3t^2 - 1 \qquad\qquad \frac{dy}{dt} = -2t$$

From the chain rule we have

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ &= \frac{-2t}{3t^2 - 1} \end{aligned}$$

Example

Suppose we wish to find $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when

$$x = t^3 + 3t^2 \qquad y = t^4 - 8t^2$$

Differentiating

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 3t^2 + 6t \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = 4t^3 - 16t$$

Then, using the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}t}}{\frac{\mathrm{d}x}{\mathrm{d}t}} \qquad \text{provided} \quad \frac{\mathrm{d}x}{\mathrm{d}t} \neq 0$$

so that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4t^3 - 16t}{3t^2 + 6t}$$

This can be simplified as follows

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{4t(t^2-4)}{3t(t+2)} \\ &= \frac{4t(t+2)(t-2)}{3t(t+2)} \\ &= \frac{4(t-2)}{3}\end{aligned}$$

We can apply the chain rule a second time in order to find the second derivative, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) \\ &= \frac{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)}{\frac{\mathrm{d}x}{\mathrm{d}t}} \\ &= \frac{\frac{4}{3}}{3t^2 + 6t} \\ &= \frac{4}{9t(t+2)}\end{aligned}$$

DIFFERENTIATION FORMULA

1. $\frac{d}{dx}[x] = 1$
2. $\frac{d}{dx}\left[\frac{x^{r+1}}{r+1}\right] = x^r \quad (r \neq -1)$
3. $\frac{d}{dx}[\sin x] = \cos x$
4. $\frac{d}{dx}[-\cos x] = \sin x$
5. $\frac{d}{dx}[\tan x] = \sec^2 x$
6. $\frac{d}{dx}[-\cot x] = \csc^2 x$
7. $\frac{d}{dx}[\sec x] = \sec x \tan x$

DIFFERENTIATION FORMULA

8. $\frac{d}{dx}[-\csc x] = \csc x \cot x$
9. $\frac{d}{dx}[e^x] = e^x$
10. $\frac{d}{dx}\left[\frac{b^x}{\ln b}\right] = b^x \quad (0 < b, b \neq 1)$
11. $\frac{d}{dx}[\ln |x|] = \frac{1}{x}$
12. $\frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$
13. $\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$
14. $\frac{d}{dx}[\sec^{-1} |x|] = \frac{1}{x\sqrt{x^2-1}}$